

SIMULATION CONDITION

Parameter	Description	Specification	Unit
V_{in}	Input voltage	1.2	V
V_{out}	Output voltage	0.8	V
V_{ref}	Reference voltage from Bandgap	0.6	V
VSS	Min supply voltage	0	V
R_{load}	Resistor loading	16 @ high load 8000 @ low load	Ω
Temp	Simulation temperature	25	degree
Corner, Tech		Typical, tsmc 65nm	
Device type		pch/nch_lvt	

Parameter	Description	Specification	Allowable range
R	Resistor value	Can use ideal resistors from analog lib	$1\Omega - 500k\Omega$
W_{fg}	Finger width	$fg * 0.5 * 120n$ <i>fg is an integer variable</i>	$900 \geq fg \geq 2$
Multi	Multiplier	m <i>m is an integer variable</i>	$300 \geq m \geq 1$
L	Channel length	$stk * 60n$ <i>stk is an integer variable</i>	$300 \geq stk \geq 1$

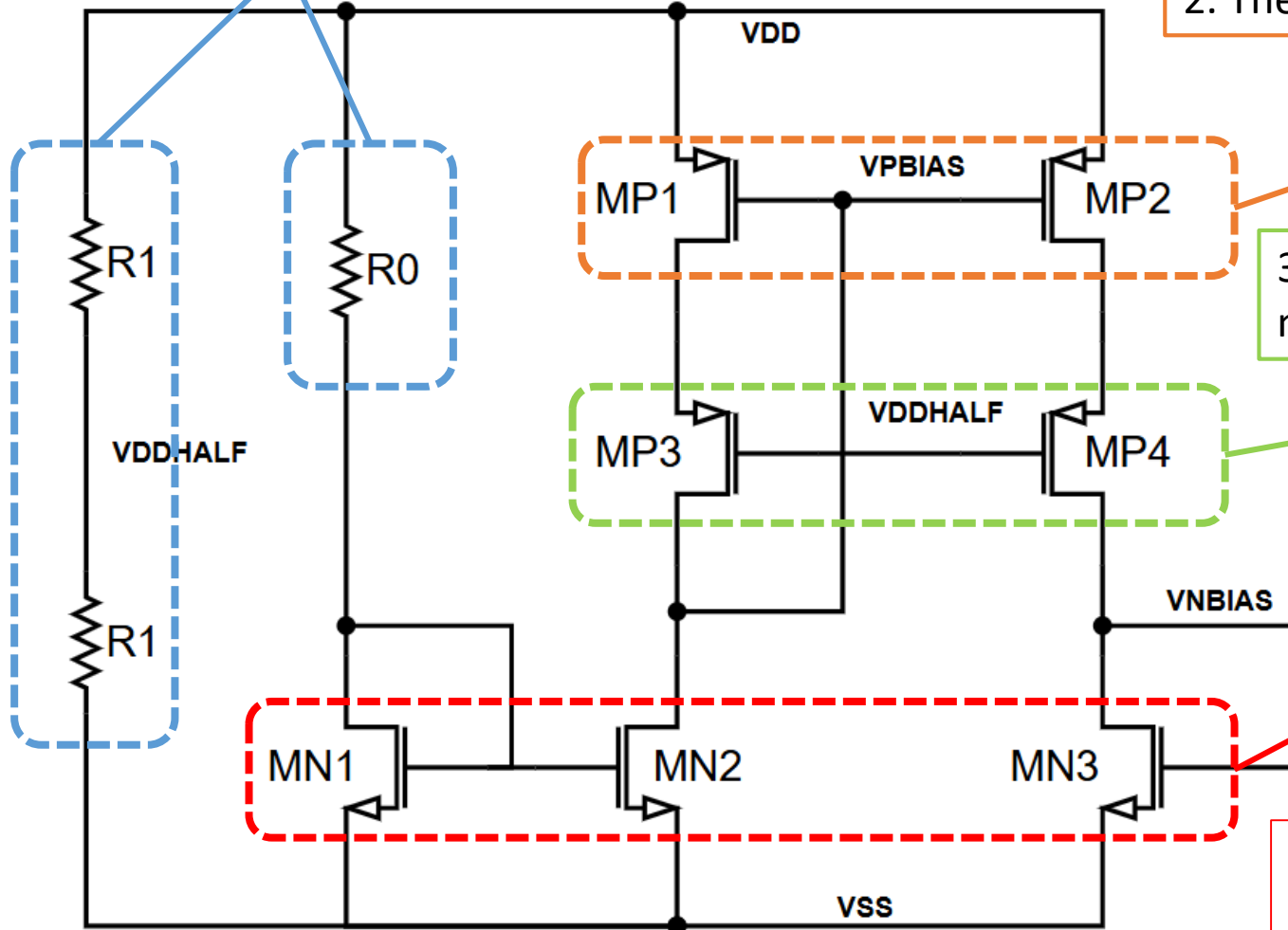
****** only apply $m \geq 1$ for PPASS,
other devices have $m = 1$

LDO SPECIFICATION

Testbench	Parameter	Description	Specification	Unit
DC	V_{od}	Overdrive voltage ($V_{gs} - V_{th}$)	≥ -100 for all mosfet	mV
	V_{margin}	Margin voltage ($V_{ds} - V_{dsat}$)	≥ 0 for all mosfet	mV
Transient	$V_{out,err}$	Output voltage error	± 8	mV
	LNR	Line regulation, $\pm 10\%$ variation of V_{in}	≤ 8	mV/V
	LDR	Load regulation, i_{load} varies from max to min	≤ 40	mV/A
AC	f_{UGB}	Unity Gain Bw	≥ 5	MHz
	A_{CLG}	DC Loop Gain	≥ 50	dB
	PM	Phase Margin of loop	≥ 60	$degree$
	$PSRR$	Power Supply Rejected Ratio	$\geq 50 @ 10kHz$ $\geq 30 @ 1MHz$	dB

BIAS

- 1.1 Should choose $R1 \sim k\Omega$ to reduce static current as much as possible (recommend: **30k Ω**)
- 1.2 $R0$ should be chosen around $\sim k\Omega$ (**1-500k Ω**)



2. These 2 devices must have the same dimension (fg & stk).

3. These 2 devices must be the same (fg & stk), no need to be as same as (2)

4. They need to have specific ratios b/w each pair

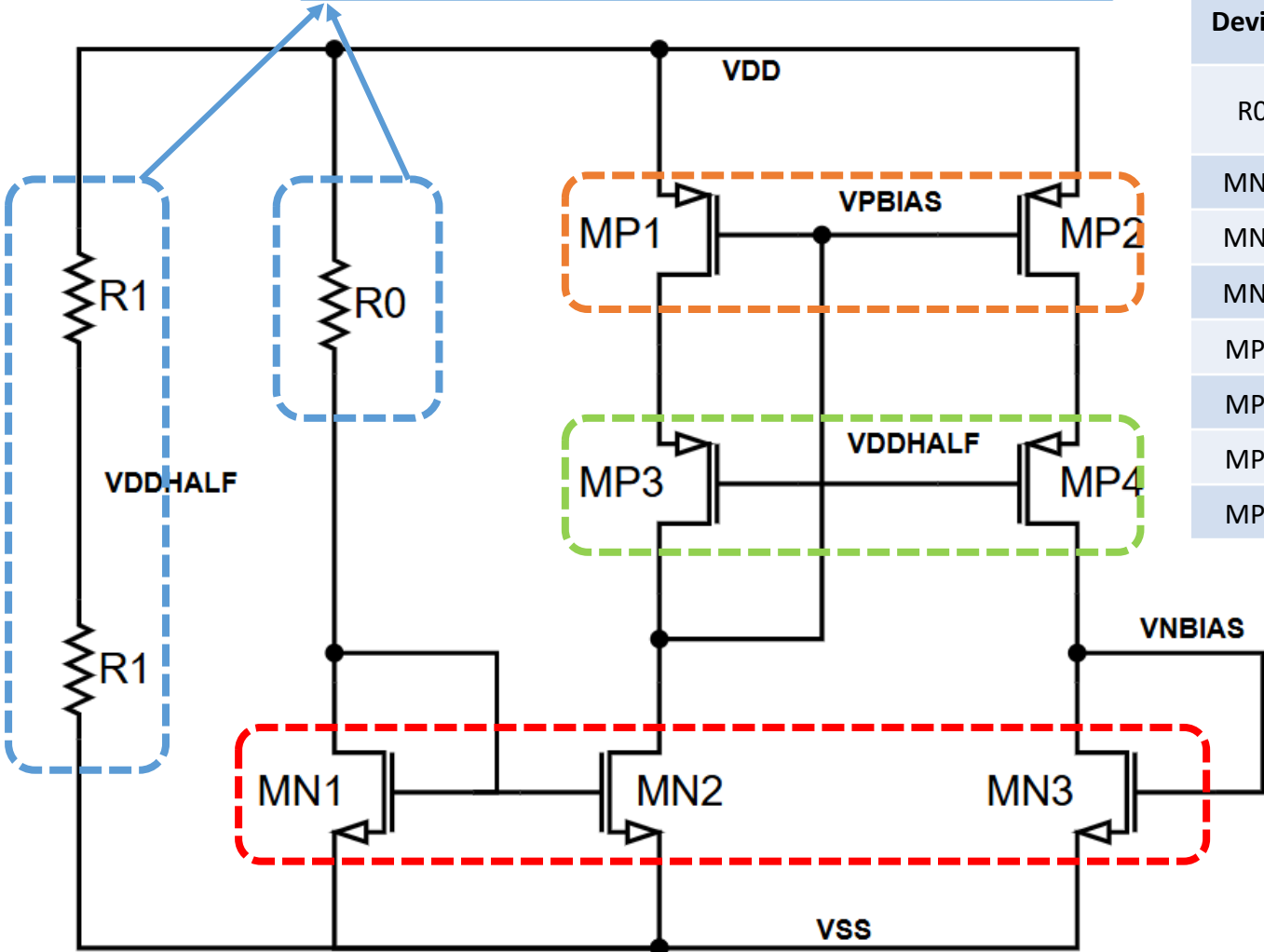
$$k \left(\frac{W}{L} \right)_{n1} = m \left(\frac{W}{L} \right)_{n2} = m \left(\frac{W}{L} \right)_{n3}$$

And k, m must be integer numbers

$$\frac{W}{L} = \frac{fg \times 0.5 \times 120nm}{stk \times 60nm} = \frac{fg}{stk}$$

BIAS

- 1.1 Should choose $R1 \sim k\Omega$ to reduce static current as much as possible (recommend: $30k\Omega$)
- 1.2 $R0$ should be chosen around $\sim k\Omega$ ($1-500k\Omega$)

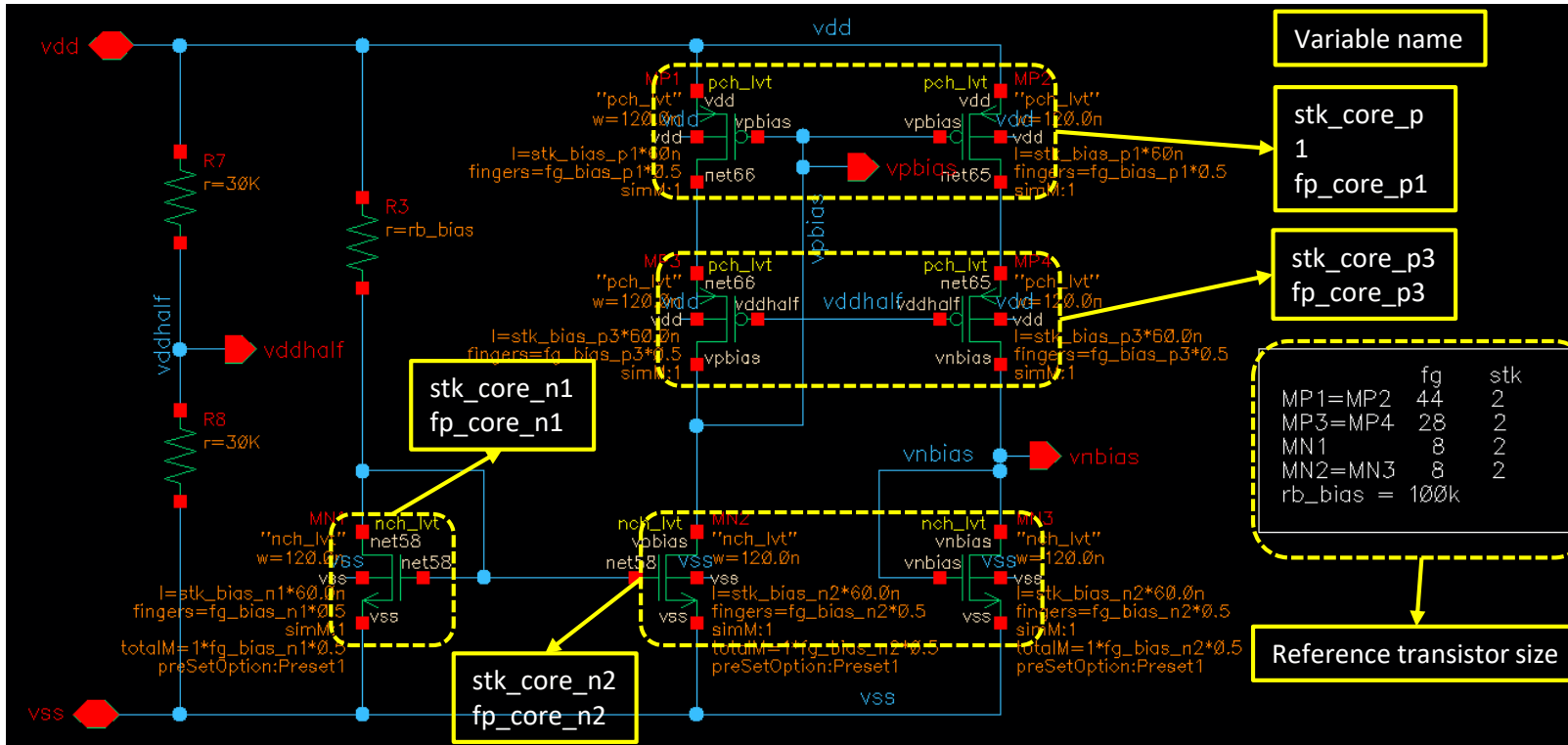


- **Note:**
- Resistance of $R0$ is $(50 \times rb) (\Omega)$
 - $R_1 = 30k\Omega \rightarrow$ fixed value

Constraint of variable names				
Device	L (m)	Number of finger	Multiplier	Range of value
R0	$50\Omega \times rb (\Omega)$			$1 \leq rb \leq 10000$ <i>rb is integer</i>
MN1	stk_bias_n1*60n	$fg_bias_n1*0.5$	1	$2 \leq fg_bias_* \leq 900$ $1 \leq stk_bias_* \leq 300$ <i>stk_bias_* and fg_bias_* are integer</i>
MN2	stk_bias_n2*60n	$fg_bias_n2*0.5$	1	
MN3			1	
MP1	stk_bias_p1*60n	$fg_bias_p1*0.5$	1	
MP2			1	
MP3	stk_bias_p3*60n	$fg_bias_p3*0.5$	1	
MP4			1	

BIAS - example

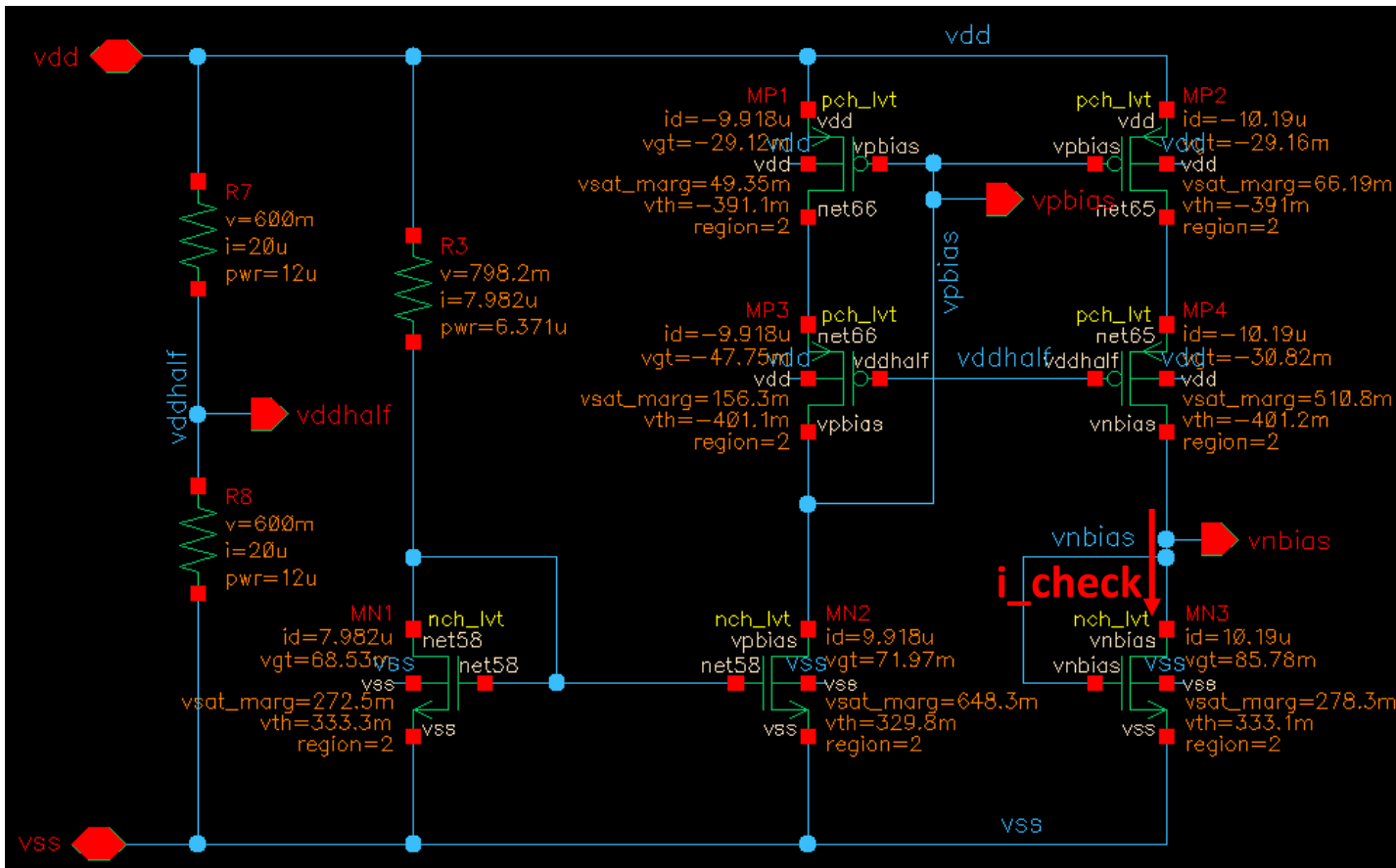
Total length = $stk * 60n$ (stk is integer number (1, 2, ...))



CDF Parameter	Value	Display
Model name	pch_lvt	off
description	CORE LOW VT PMOS transistor	off
I (M)	stk_bias_p1*60n M	off
w (M)	120.0n M	off
total_width(M)	1.2e-07*fg_bias_p1*0.5 M	off
Number of Fingers	fg_bias_p1*0.5	off
Multiplier	1	off
total_m	1*fg_bias_p1*0.5	off
Hard_constraint	☒	off
Calc Diff Params	☐	off
Source_area	8.1e-14	off
Drain_area	8.1e-14	off
Source_periphery(M)	1.8u M	off

**** must follow variable name when applying optimization algorithms**

BIAS - example



DCOP check:

1. Static current through 1st branch $\sim 20\mu A$.
2. Base current of BIAS circuit is $10\mu A$.
3. All mosfets operate in saturation/sub-threshold region
4. DCOP condition:
 1. **Region** = 2 (saturation) or 3 (sub-threshold)
 2. **Overdrive voltage** ($|V_{gs}| - |V_{th}| \geq -100mV$)
 3. **Voltage margin** ($|V_{ds}| - |V_{dsat}| \geq 0mV$)

How to check? → DCOP verification sequence

1. If region = 2 → **PASS DCOP**
2. If region = 3 → check $|V_{gt}| \leq 100 mV$ or not?
 If $|V_{gt}| \leq 100 mV$ → **PASS DCOP**
 Else → **FAIL DCOP** check

We have: $i_{check} = id_{mn3}$ → we need this value around $10\mu A$ as base current
 → Allowable value range: $[9.5 ; 10.5] (\mu A)$

Testbench	Parameter	Description	Specification	Unit
DC	<i>region</i>	Operation region	2 or 3 (follow the verification sequence above)	
	<i>i_{check}</i>	Base current	$9.5 \leq i_{check} \leq 10.5$	μA

BIAS - example

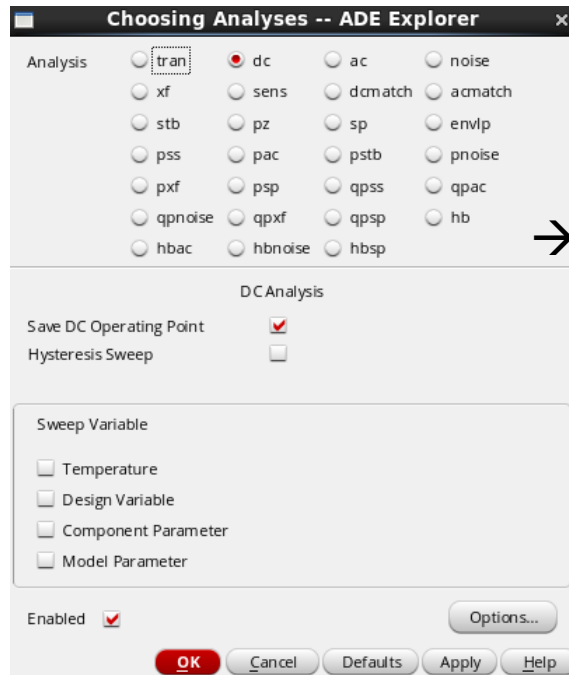
Name	Type	Details	Value	Plot	Save	Spec
region_bias_mn1	expr	OP("/bias/MN1" "region")	2	☒	☐	
vgt_bias_mn1	expr	abs(OP("/bias/MN1" "vgt"))	68.53m	☒	☐	
vmarg_bias_mn1	expr	OP("/bias/MN1" "vsat_marg")	272.5m	☒	☐	
id_bias_mn1	expr	OP("/bias/MN1" "id")	7.982u	☒	☐	
id_bias_mn3	expr	OP("/bias/MN3" "id")	10.19u	☒	☐	
icheck_err	expr	id_bias_mn3	10.19u	☒	☐	range 9.5u 10.5u

Name of parameters

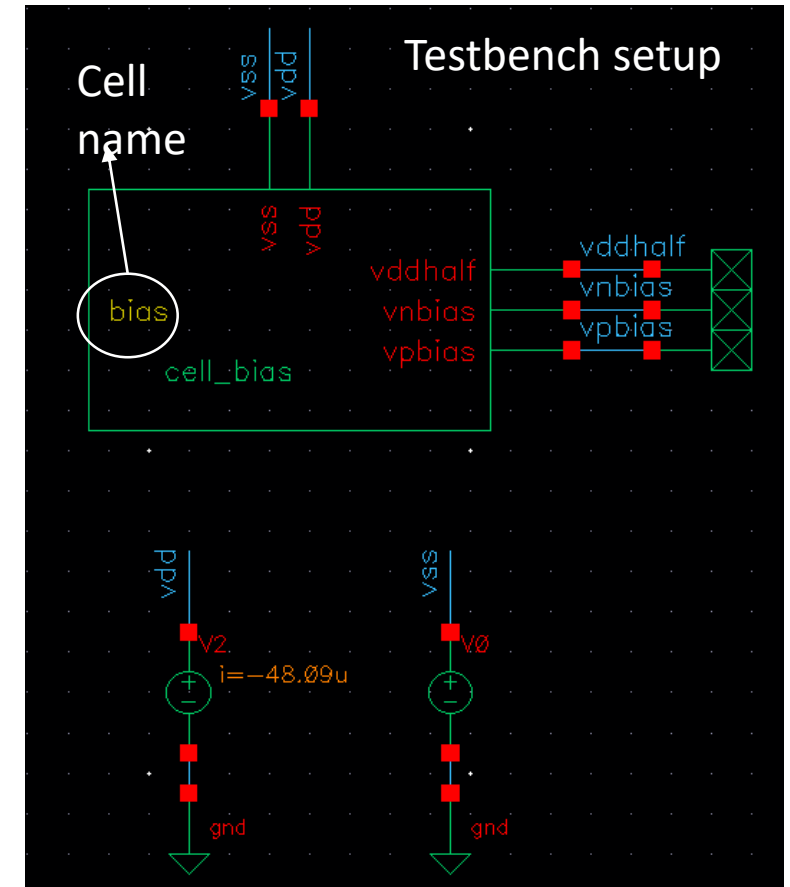
Measurement → can include in OCEAN file to do the simulation

i_{check} measurement

Example of DCOP check of 1 device → need to check all devices in schematic



→ DCOP analysis



Core

** When we separately design EA, there are no more information about **core voltage** from core cell → replace core devices by current sources and just need to evaluate the optimized core current

1. Must be the same (fg & stk)

$I_{bp}=i_0$

$I_{bp}=i_0$

2. Replace 2 identical core PMOS by 2 ideal current source having same current value

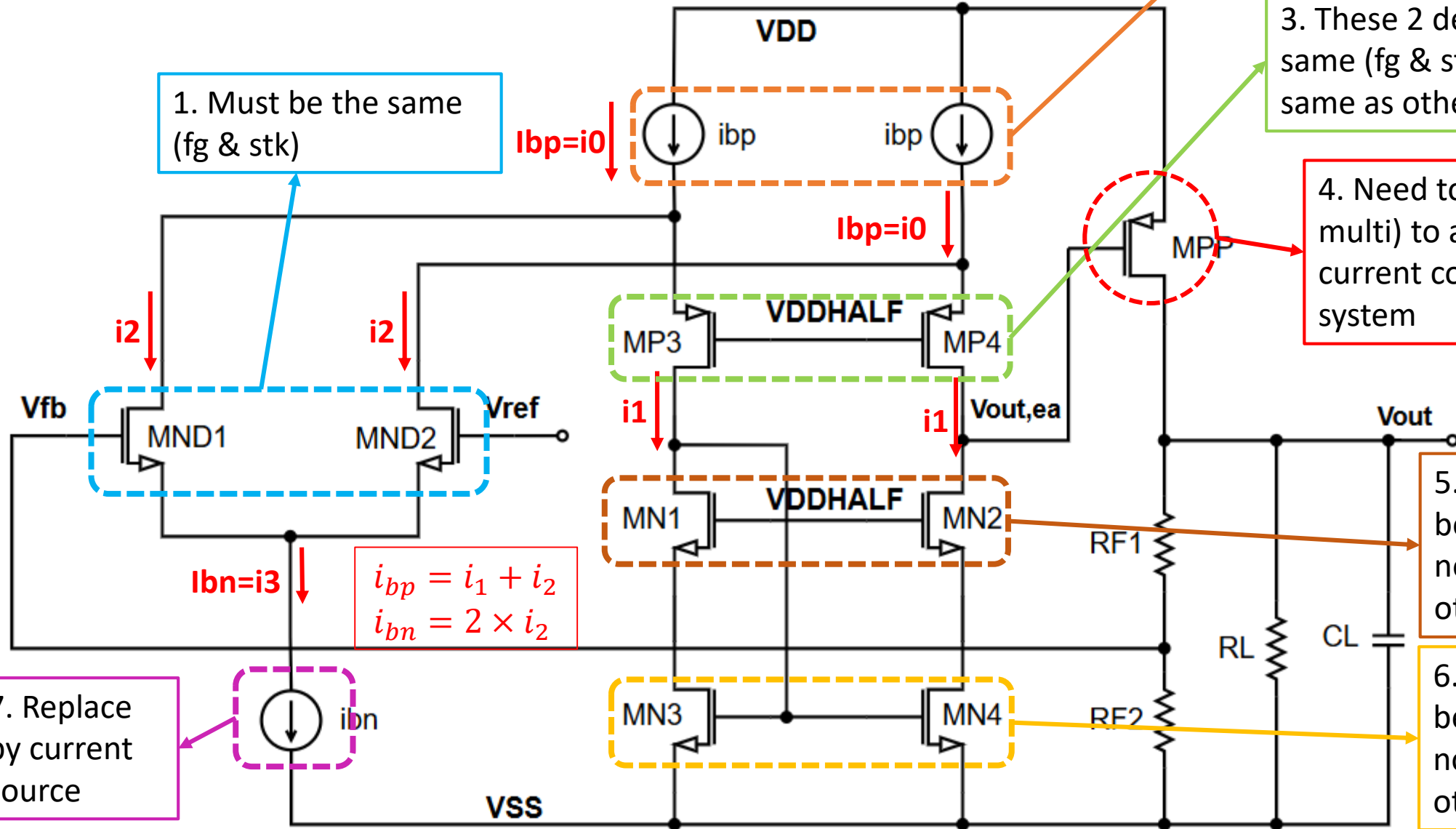
3. These 2 devices must be the same (fg & stk), no need to be as same as other pair

4. Need to choose (fg, stk & m – multi) to adapt both min & max current consumption from load system

5. These 2 devices must be the same (fg & stk), no need to be as same as other pair

6. These 2 devices must be the same (fg & stk), no need to be as same as other pair

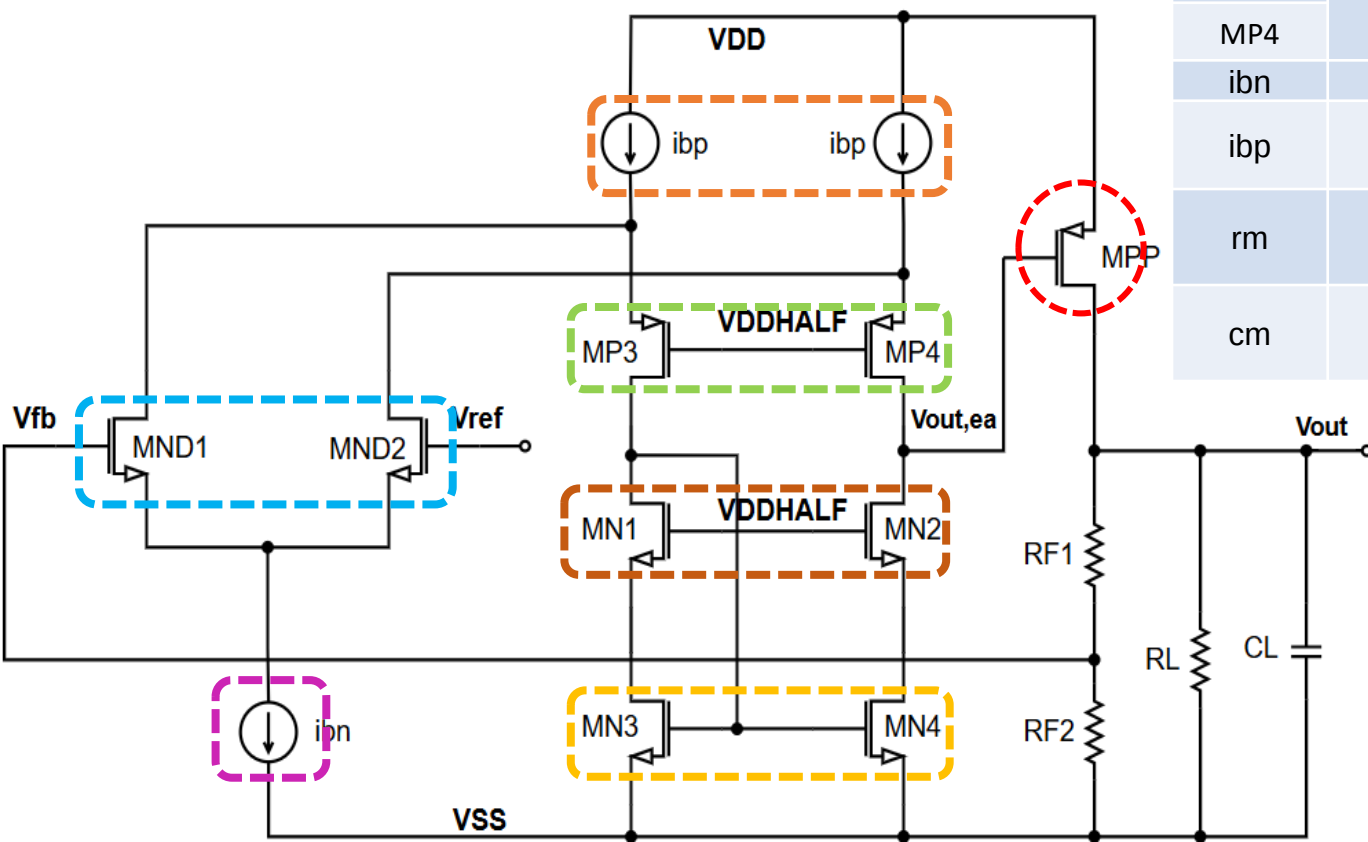
7. Replace by current source



Core

**Note:

- Current of isource (ibn) is $5 \times \mathbf{ibn} (\mu A)$
- Current of isource (ibp) is $5 \times \mathbf{ibp} (\mu A)$
- Rm, cm connection shows in next slide



Constraint of variable names				
Device	L (m)	Number of finger	Multiplier	Range of value
MPP	stk_core_mpp*60n	fg_core_mpp*0.5	m_core_mpp	$2 \leq fg_core_* \leq 900$ $1 \leq stk_core_* \leq 300$ $1 \leq m_core_* \leq 300$ <i>stk_core_*, fg_core_*</i> <i>and fg_core_* are integer</i>
MND1	stk_core_nd1*60n	fg_core_nd1*0.5	1	
MND2			1	
MN1	stk_core_n1*60n	fg_core_n1*0.5	1	
MN2			1	
MN3	stk_core_n3*60n	fg_core_n3*0.5	1	
MN4			1	
MP3	stk_core_p3*60n	fg_core_p3*0.5	1	
MP4			1	
ibn	$5\mu A \times ibn$, ibn is integer			$1 \leq ib^* \leq 1000$
ibp	$5\mu A \times ibp \geq \left(\frac{5\mu A \times ibn}{2} + 5\mu A \right)$, ibp is integer			
rm	$50\Omega \times rm (\Omega)$			$1 \leq rm \leq 10000$ <i>rm is integer</i>
cm	$100fF \times cm (fF)$			$1 \leq cm \leq 500$ <i>rm is integer</i>

****Note:**

- $RF1 = 10k\Omega; RF2 = 30k\Omega$
- $C_L = 100pF$

Core - example

Total length = $stk * 60n$ (stk is integer number (1, 2, ...))

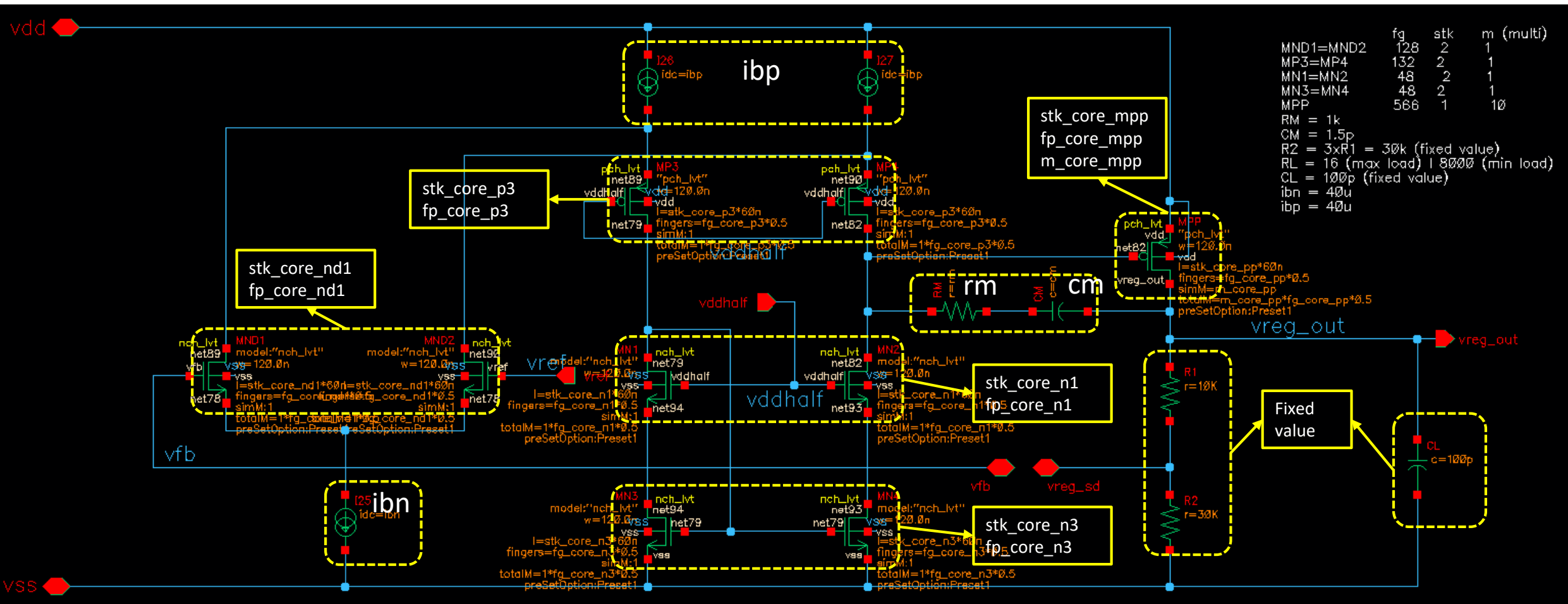
Min finger width

Automatically adjust

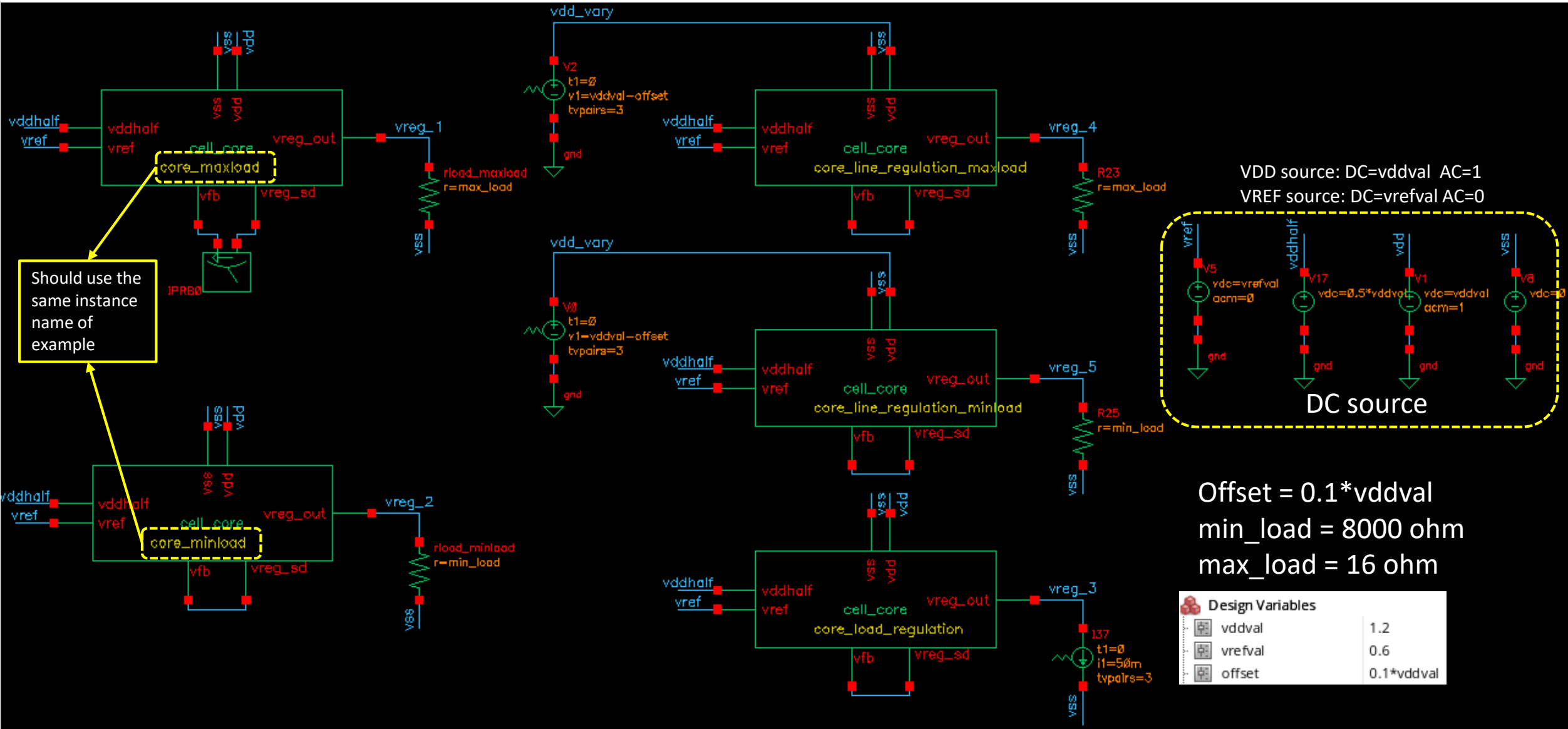
n.o fingers, *0.5 means we want 60n per step

Keep multi=1, except MPP

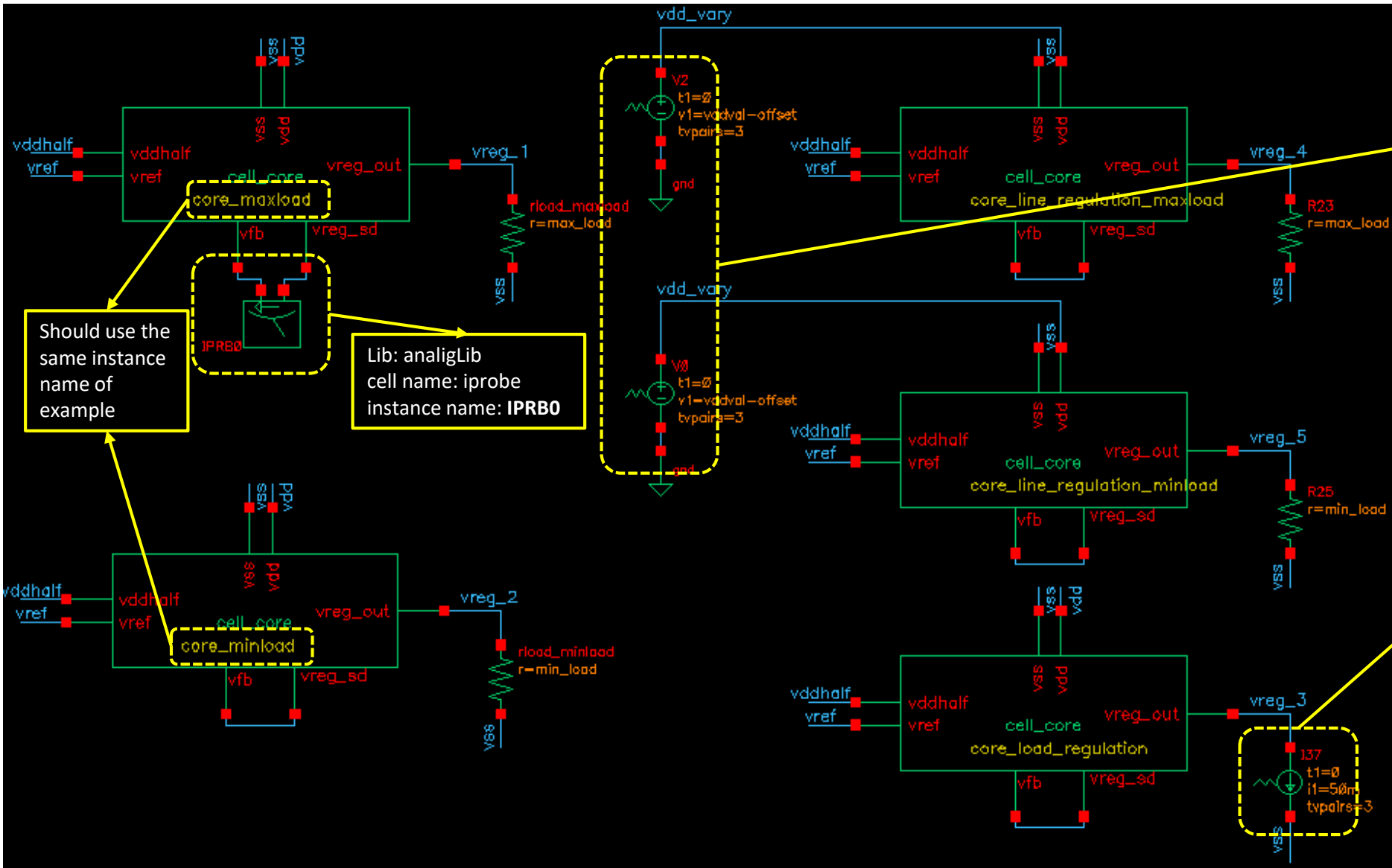
$stk_core_pp * 60n$	$120.0n$	$1.2e-07 * fg_core_pp * 0.5$	$fg_core_pp * 0.5$	m_core_pp	$m_core_pp * fg_core_pp * 0.5$
$w(M)$	$total_width(M)$	Number of Fingers	Multiplier	$total_m$	



Core - testbench



Core - testbench



Should use the same instance name of example

Lib: analogLib
cell name: iprobe
instance name: IPRB0

Property	Value	Display
Library Name	analogLib	off
Cell Name	vpw1	off
View Name	symbol1	off
Instance Name	V2	off

Buttons: Add, Delete, Modify

User Property	Master Value	Local Value	Display
Ivsignore	TRUE		off

CDF Parameter

Parameter	Value	Display
Number of pairs of points	3	off
Time 1	0 s	off
Voltage 1	vddval-offset V	off
Time 2	10n s	off
Voltage 2	vddval-offset V	off
Time 3	10.01n s	off
Voltage 3	vddval+offset V	off
DC voltage		off
AC magnitude		off
AC phase		off

Buttons: Browse, Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	ipw1	off
View Name	symbol1	off
Instance Name	I37	off

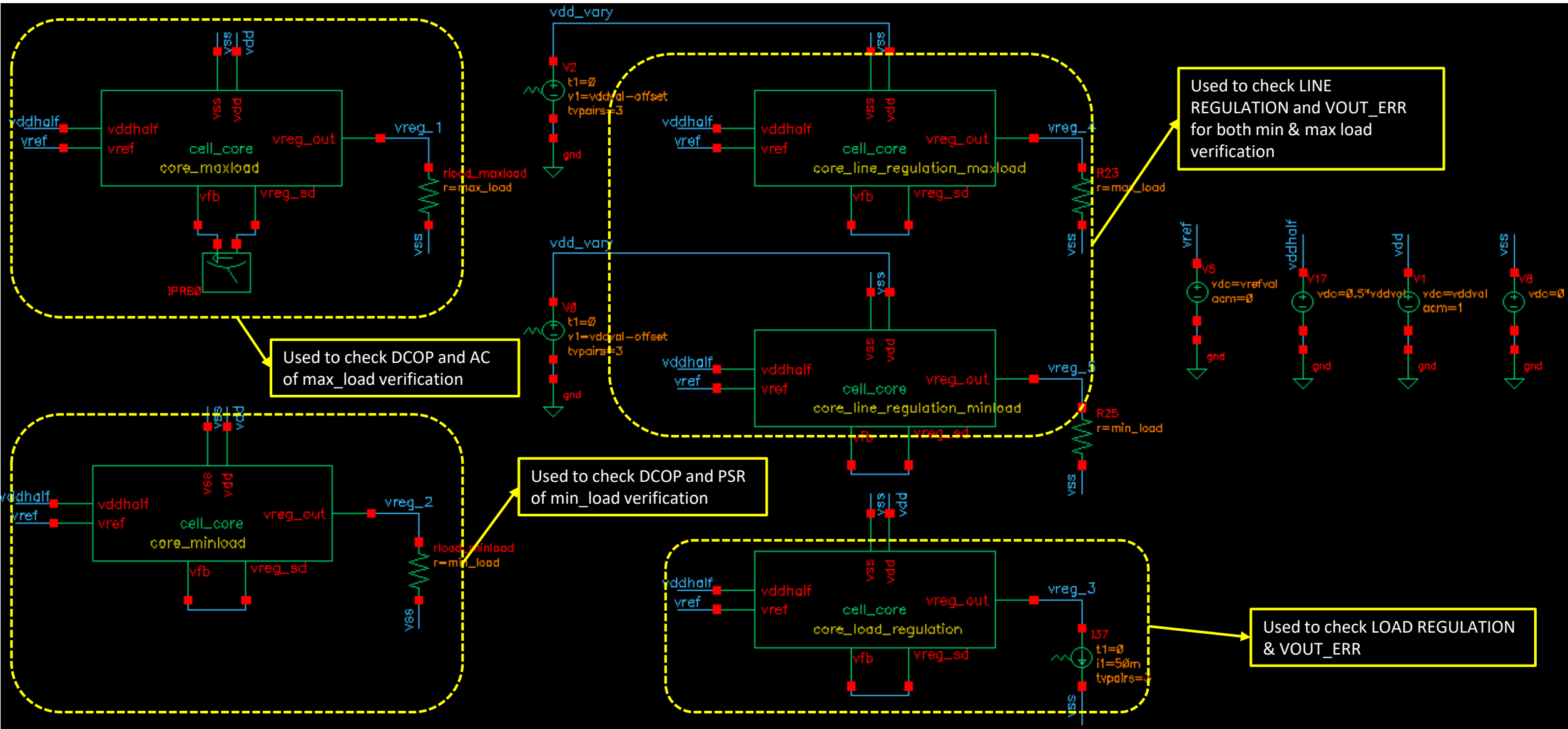
Buttons: Add, Delete, Modify

User Property	Master Value	Local Value	Display
Ivsignore	TRUE		off

CDF Parameter

Parameter	Value	Display
Number of pairs of points	3	off
Time 1	0 s	off
Current 1	50m A	off
Time 2	10n s	off
Current 2	50m A	off
Time 3	10.01n s	off
Current 3	100u A	off
DC current		off
AC magnitude		off

Core - testbench



Core - testbench

Testbench	Item	Parameter	Measurement	Unit
DCOP	Region	region_core_nd1	OP("/core_maxload/MND1" "region")	
	V_{od}	vod_core_nd1	abs(OP("/core_maxload/MND1" "vgt"))	V
	V_{margin}	vmarg_core_nd1	OP("/core_maxload/MND1" "vsat_marg")	V
Transient	Line regulation	vout_vinmax_maxload	value(vtime('tran "/vreg_4") 9.98e-07)	V
		vout_vinmin_maxload	value(vtime('tran "/vreg_4") 8e-09)	V
		LNR_maxload	((abs((vout_vinmax_maxload - vout_vinmin_maxload)) / 0.24) * 1000)	mV/V
		vout_vinmax_minload	value(vtime('tran "/vreg_5") 9.98e-07)	V
		vout_vinmin_minload	value(vtime('tran "/vreg_5") 8e-09)	V
		LNR_minload	((abs((vout_vinmax_minload - vout_vinmin_minload)) / 0.24) * 1000)	mV/V
		LNR	max(LNR_maxload, LNR_minload)	mV/V
	Load regulation	vout_minload	value(vtime('tran "/vreg_5") 9.98e-07)	V
		vout_maxload	value(vtime('tran "/vreg_5") 8e-09)	V
		LDR	((abs((vout_minload - vout_maxload)) / 0.0499) * 1000)	mV/V

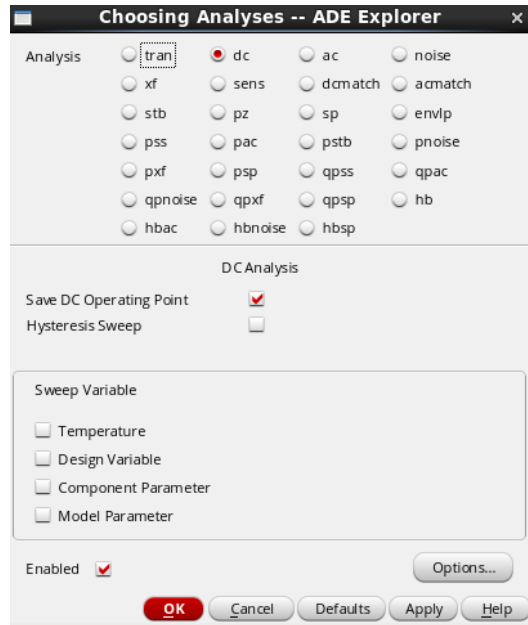
Core - testbench

Testbench	Item	Parameter	Measurement	Unit
AC	f_{UGB}	UGB	cross(db(mag(getData("loopGain" ?result "stb"))) 0 1 "either" nil nil)	MHz
	A_{CLG}	Av	value(db(mag(getData("loopGain" ?result "stb"))) 1)	dB
	PM	PM	value(phaseDegUnwrapped(getData("loopGain" ?result "stb")) cross(db(mag(getData("loopGain" ?result "stb"))) 0 1 "either" nil nil))	degree
	PSRR	PSR_10k_maxload	abs(value(db(vfreq('ac "/vreg_1")) 10000))	dB
		PSR_1meg_maxload	abs(value(db(vfreq('ac "/vreg_1")) 1000000))	dB
		PSR_10k_minload	abs(value(db(vfreq('ac "/vreg_2")) 10000))	dB
		PSR_1meg_minload	abs(value(db(vfreq('ac "/vreg_2")) 1000000))	dB

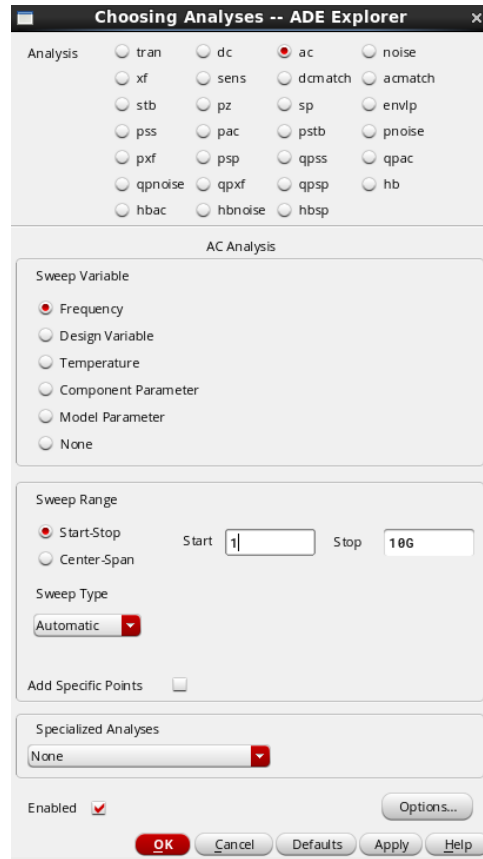
Example:

Av	expr	value(db(mag(getData("loopGain" ?result "stb"))) 1)	47.71	☒	☐	> 50
UGB	expr	cross(db(mag(getData("loopGain" ?result "stb"))) 0 1 "either" nil nil)	18.69M	☒	☐	> 20M
PM	expr	value(phaseDegUnwrapped(getData("loopGain" ?result "stb")) cross(db(mag(getData("loopGain" ?result "stb"))) 0 1 "either" nil nil))	95.61	☒	☐	> 60
PSR_10k_maxload	expr	abs(value(db(vfreq('ac "/vreg_1")) 10000))	41.21	☒	☐	> 50
PSR_1M_maxload	expr	abs(value(db(vfreq('ac "/vreg_1")) 1000000))	27.32	☒	☐	> 30

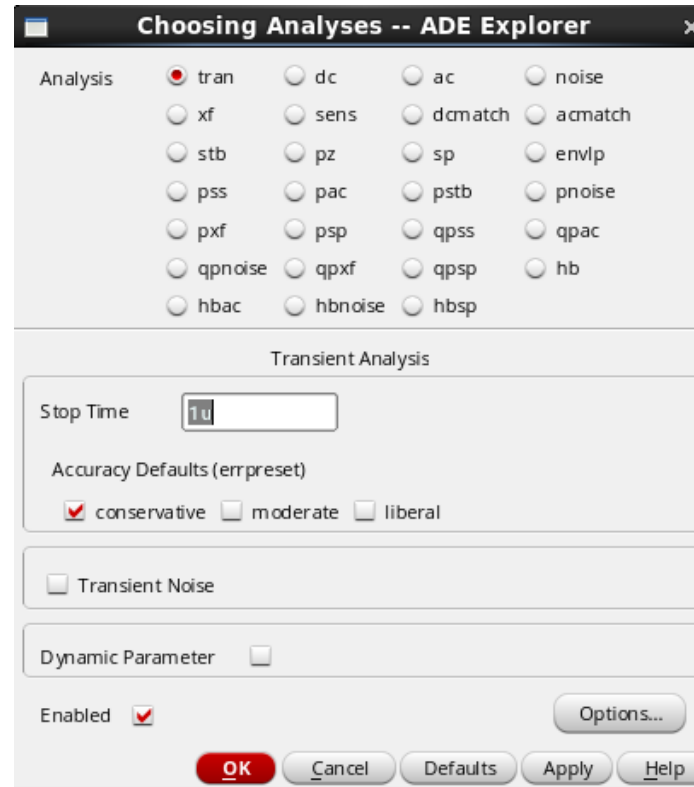
Core - testbench



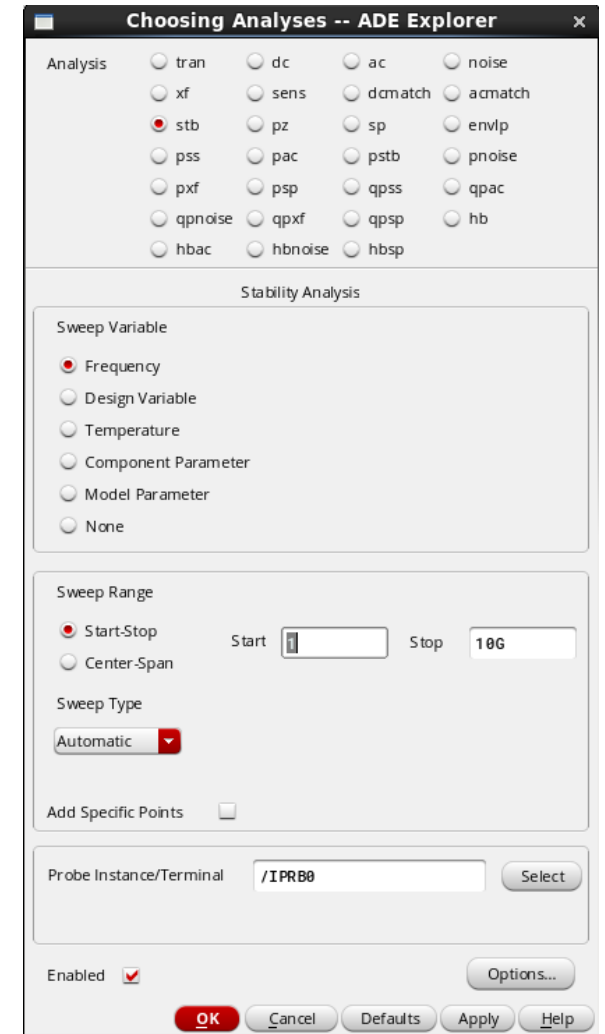
DCOP analysis



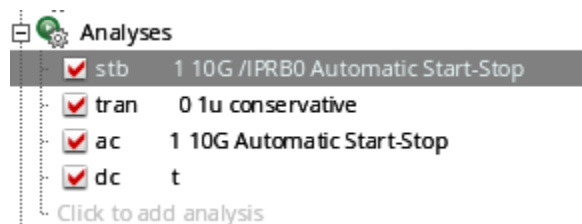
AC analysis



TRANSIENT analysis



STB analysis



→ Execute these analyses in parallel

Core Specification

Testbench	Parameter	Description	Specification	Unit
DC	V_{od}	Overdrive voltage ($V_{gs} - V_{th}$)	≥ -100 for all mosfet	mV
	V_{margin}	Margin voltage ($V_{ds} - V_{dsat}$)	≥ 0 for all mosfet	mV
Transient	$V_{out,err}$	Output voltage error (max & min load)	± 8	mV
	LNR	Line regulation, $\pm 10\%$ variation of V_{in} (min & max load)	≤ 8	mV/V
	LDR	Load regulation, iload varies from max to min	≤ 40	mV/A
AC	f_{UGB}	Unity Gain Bw	≥ 5	MHz
	A_{CLG}	DC Loop Gain	≥ 50	dB
	PM	Phase Margin of loop	≥ 60	$degree$
	$PSRR$	Power Supply Rejected Ratio	$\geq 50 @ 10kHz$ $\geq 30 @ 1MHz$	dB

Objective Functions for LDO

Bias Circuit

After pass all the requirement (DCOP, i_check)

$$FOM = 1 - \text{abs} \left(\frac{i_{\text{check}}(\mu A)}{10\mu A} - 1 \right)$$

$\rightarrow$ **FOM higher is better**

Core

After pass all the requirement (DCOP, Trans, AC)

$$FOM = \frac{\Delta V_{out}}{V_{out}} \times \frac{I_{in} - I_{load,max}}{I_{load,max}} \times \frac{1}{PSR@1MHz}$$

$\rightarrow$ **FOM smaller is better**

Note:

- $\Delta V_{out} = V_{out,max} - V_{out,min}$ (V); $V_{out} = 0.8V$
- $I_{in} = i_{sub}(V_{in})$ in DC analysis of MAX_LOAD verification (A) $\rightarrow$ total current consumption of LDO; $I_{load,max} = 0.05$ (A)
- $PSR@1MHz \rightarrow$ abs value of PSR at $f=1MHz$ in linear scale: $PSR_{linear} = 10^{PSR_{dB}/20}$